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7. *(a)* Group 7 elements are known as the halogens. The following table shows the observations made when the first three members of the group react with hydrogen.

Halogen	Observations
fluorine	explodes in cold and dark
chlorine	explodes in sunlight
bromine	small explosion when ignited with a flame

(i) Use your knowledge of electronic structure to explain why all the halogens react in a similar way and why they react more slowly on going down the group. [3]

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(ii) Hydrogen fluoride is highly corrosive and can be used to etch glass which is mainly silicon dioxide.

Balance the symbol equation for the reaction between hydrogen fluoride and silicon dioxide. [1]

$$\square\ HF + SiO_2 \longrightarrow SiF_4 + \square\ H_2O$$

(iii) Calcium fluoride reacts with sulfuric acid, H_2SO_4, to produce calcium sulfate and hydrogen fluoride. Give the **symbol** equation for the reaction. [3]

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(b) Chlorine reacts with aluminium to produce aluminium chloride.

A sample of aluminium chloride of mass 26.70 g was found to contain 5.45 g of aluminium. Calculate the simplest formula of this chloride of aluminium.

You **must** show your working. [3]

$A_r(\mathrm{Al}) = 27 \qquad A_r(\mathrm{Cl}) = 35.5$

Formula ....

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